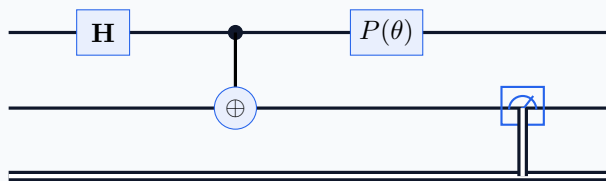


# QPU Documentation Suite

[Start Here](#)



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This short index points to four focused guides: a beginner walkthrough, classical and quantum theory, the complete QPU language reference, and worked examples with troubleshooting.

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# 1 Choose the guide you need

The documentation is split so a new user does not need to read a large reference manual from beginning to end.

## 1.1 New to programming or the website

### Open QPU Getting Started

Begin here to learn what source text, instructions, arguments, compilation, wires, tokens, and gates mean. It gives an exact website workflow and complete protocols that can be copied into the editor.

## 1.2 New to classical or quantum theory

### Open Classical and Quantum Theory for QPU

This guide starts with bits, Boolean algebra, and truth tables, then develops complex numbers, Euler's formula, qubits, the Bloch sphere, tensor products, entanglement, interference, measurement, and the mathematics of every gate.

## 1.3 Looking up exact syntax

### Open QPU Language Reference

Use this guide for token and constant forms, flags, gate arity, every supported instruction, child-process binding, truth-table file syntax, compatibility behavior, errors, grammar, and the glossary.

## 1.4 Learning through examples or fixing an error

### Open QPU Examples and Troubleshooting

This guide explains a first circuit, reversible half-adder, interference experiment, child-process example, common protocol errors, diagnostic codes, and when to open the Correction Lab.

# 2 Recommended learning path

Getting Started explains each step in detail, with a starter circuit and a checkpoint question. The starter-circuit cards in the Circuit builder are numbered with the same steps.

| Step                       | What you learn                                                        | Where to read                                                     |
|----------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>1. Single qubit</b>     | X, H, Z, and measurement; superposition versus interference.          | Getting Started; Theory Guide (single-qubit gates)                |
| <b>2. Controlled gates</b> | CNOT, CZ, CY, SWAP: which wire is read and which is changed.          | Theory Guide (controlled gates)                                   |
| <b>3. Reversible logic</b> | CCNOT and AND/OR/XOR/NAND: inputs are kept, answers go into a target. | Getting Started (half-adder); Theory Guide (derived gates)        |
| <b>4. Entanglement</b>     | Bell states and phase kickback.                                       | Theory Guide (multiple qubits); Examples (phase kickback)         |
| <b>5. Arithmetic</b>       | Full adders, multi-bit adders, comparators, parity.                   | Examples (advanced worked circuits)                               |
| <b>6. Multi-stage</b>      | Child processes, uncomputation, oracles, custom gates.                | Language Reference (process composition); Examples (custom gates) |

Throughout, keep the Language Reference open while authoring a protocol, and use Protocol checks and the Correction Lab when compilation or truth-table tests report a problem.

### 3 Find the explanation for what is on screen

- **A button or selector:** Getting Started, *Every control on screen, by its exact name.*
- **A gate:** select it in the workbench and open *Read the full gate reference*, or see the Theory Guide gate chapters.
- **Reset state, Clear circuit, or Reset site:** *Reset: an operation that is not a gate*, in Getting Started, the Theory Guide, and the Language Reference.

### 4 File map

| PDF                                  | Purpose                                                      |
|--------------------------------------|--------------------------------------------------------------|
| QPU_Getting_Started.pdf              | Beginner programming concepts and website use.               |
| QPU_Theory_Guide.pdf                 | Classical/quantum theory and gate mathematics.               |
| QPU_Language_Reference.pdf           | Complete syntax and behavior reference.                      |
| QPU_Examples_and_Troubleshooting.pdf | Runnable examples, diagnostics, and Correction Lab guidance. |

### 5 A safe first action

Open the Circuit builder, load a starter circuit, choose **Play Sequence**, and observe each gate before writing source. Nothing is uploaded to a server for simulation; the state-vector simulator runs in the browser.

When ready to edit text, use a listing explicitly marked **Complete runnable protocol**. Instruction fragments are reference material and may not compile by themselves.